# Notes: Lane (QJE, 2025)

## Manufacturing Revolutions: Industrial Policy and Industrialization in South Korea

## Contents

# 1 Executive summary

South Korea's 1973–1979 Heavy and Chemical Industry (HCI) drive provides a sharp, sector-targeted industrial policy episode. Lane (2025) combines newly digitized manufacturing census/survey data, producer prices, trade flows, and an input–output (IO) table to estimate the policy's causal effects on *industrial development* (output, value added, employment, plant counts, prices, etc.) and *export development* (revealed comparative advantage and gravity-based export productivity). The core empirical design is a dynamic difference-in-differences (event study) comparing targeted vs. non-targeted industries before and after 1973, with robustness using doubly robust DiD estimators and cross-country triple differences. The main findings are that targeted sectors expand strongly and persistently, upgrade their comparative advantage gradually (effects keep growing after the policy ends), exhibit stronger learning-by-doing (both within-plant and industry-wide spillovers), and generate downstream gains through production-network linkages (industries that use targeted intermediates expand and see lower output prices).

# 2 Institutional setting and treatment definition

## 2.1 The HCI drive as a sectoral "investment policy" shock

The HCI drive (1973–1979) is characterized by a concentrated set of interventions aimed at heavy and chemical industries, including:

- **Investment incentives** (tax benefits / accelerated depreciation / credit programs) that lower the user cost of capital in targeted sectors,
- **Directed credit** (e.g., Korea Development Bank lending) that disproportionately expands financing for targeted industries,
- **Complementary policies** such as protection and procurement, depending on sector and year.

## 2.2 Treatment indicator

Let $i$ index industries (either a more aggregated *long* four-digit panel, 1967–1986, or a more granular *short* five-digit panel, 1970–1986). Define:

$$\text{Targeted}_i = \begin{cases} 1 & \text{if industry } i \text{ appears in the HCI legislation / targeted sector list,} \\ 0 & \text{otherwise.} \end{cases}$$

The treatment assignment is policy-driven (legislation-based), not chosen to fit outcomes.

# 3 Data and key outcomes

## 3.1 Core datasets

- **Manufacturing industry panels (MMS):** Mining and Manufacturing Surveys/Census (EPB), harmonized across KSIC revisions to create a long (4-digit, 1967–1986) and short

(5-digit, 1970–1986) industry panel.

- **Producer prices:** Bank of Korea producer price indices digitized and mapped to industries.
- **Plant-level microdata (post-1979):** establishment data used for production-function-based TFP estimates.
- **IO table (1970 basic IO):** $\approx 320$ sector IO accounts digitized to construct linkage exposure measures.
- **Trade flows:** SITC rev.1 (4-digit) trade data (UN Comtrade) and policy data (tariffs/quantitative restrictions in selected years).

### 3.2 Industrial-development outcomes (examples)

Common outcomes $y_{it}$ include:

- log real output / shipments,
- log value added, employment, number of plants,
- log output prices (producer price indices),
- labor productivity log(VA per worker),
- industry shares of manufacturing (output share, labor share).

### 3.3 Export-development outcomes (examples)

**Balassa revealed comparative advantage (RCA).** For country $c$, product/industry $i$, year $t$:

$$\mathrm{RCA}_{cit} \equiv \frac{X_{cit}/\sum_k X_{ckt}}{\sum_{c'} X_{c'it}/\sum_{c',k} X_{c'kt}}. \tag{1}$$

Lane uses variants that handle zeros:

- $\log(\mathrm{RCA}_{cit})$ when defined,
- $\operatorname{asinh}(\mathrm{RCA}_{cit})$ as a smooth transform that accommodates zeros.

**Comparative-advantage indicator.**

$$\mathbb{I}\{\mathrm{RCA}_{cit} > 1\}.$$

**Gravity-based "relative export productivity (CDK)."** Following Costinot, Donaldson, and Komunjer (2012), trade flows can be used to back out a technology component for each country–product that is consistent with a Ricardian gravity structure. Lane reports a *relative export productivity* measure (denoted "CDK") to complement reduced-form RCA outcomes. (Exact implementation details are in the paper and online appendix; for interpretation, treat this as a gravity-consistent productivity proxy inferred from trade patterns.)

# 4 Model 1: Dynamic difference-in-differences event study (baseline)

## 4.1 Equation (1): dynamic DiD with industry and year fixed effects

The core specification is a dynamic DiD event study with 1972 as the omitted base year:

$$\log(y_{it}) = \alpha_i + \tau_t + \sum_{j \neq 1972} \beta_j \Big(\text{Targeted}_i \times \text{Year}_t^j\Big) + \sum_{j \neq 1972} X_i' \times \text{Year}_t^j \, \gamma_j + \varepsilon_{it}, \tag{1}$$

where:

- $\alpha_i$ are industry fixed effects,
- $\tau_t$ are year fixed effects,
- $\text{Year}_t^j$ is an indicator for calendar year $j$,
- $X_i$ are *pre-1973 industry averages* (controls) interacted with each year.

## 4.2 How to read the coefficients

- Each $\beta_j$ is the *relative difference* between targeted and non-targeted industries in year $j$, compared to 1972.
- "Parallel trends" is assessed by inspecting pre-1973 coefficients: $\{\beta_j\}_{j<1973}$ should be approximately zero (no differential pre-trend).
- Post-1973 coefficients trace out the *dynamic path* of policy effects (during the drive and after its end in 1979).

## 4.3 Mapping logs to percent effects

For outcomes in logs, a useful approximation is:

$$\%\Delta \approx 100 \cdot (\exp(\beta) - 1).$$

(The paper sometimes applies a small-sample correction using the standard error; if needed, use $100(\exp(\beta - \frac{1}{2}\widehat{SE}^2) - 1)$.)

# 5 Model 2: Doubly robust DiD (robustness / alternative estimator)

## 5.1 Motivation

A concern with classical two-way fixed effects is that differential trends may remain even after controls. Lane therefore reports a doubly robust DiD estimator that combines: (i) inverse-probability weighting (propensity-score weighting) and (ii) an outcome regression for the untreated counterfactual.

### 5.2 Equation (3): ATT for year $t$ relative to 1972

Let $\widehat{p}(X_0) = \Pr(D = 1 \mid X_0)$ be the estimated propensity score (with $D$ = Targeted) using pre-treatment covariates $X_0$, and let $\widehat{f}_{0,t}(X_0)$ be an estimated conditional outcome model for the untreated. Then Lane implements:

$$\widehat{\mathrm{ATT}}_t = \mathbb{E}\left[\Big(\frac{D}{\widehat{p}(X_0)} - \frac{1-D}{1-\widehat{p}(X_0)}\Big)\Big((Y_t - Y_{1972}) - (\widehat{f}_{0,t}(X_0) - \widehat{f}_{0,1972}(X_0))\Big)\right], \tag{3}$$

with bootstrapped standard errors.

### 5.3 Interpretation

- "Doubly robust" means consistency holds if *either* the propensity-score model *or* the outcome model is correctly specified (not necessarily both).
- This estimator targets the same object as the event-study DiD: the average treatment effect on treated industries (ATT) in each year relative to 1972.

## 6 Model 3: Cross-country triple differences in trade (international benchmark)

### 6.1 Purpose

The industry DiD uses only Korean manufacturing industries. To benchmark against global changes in demand/technology, the paper estimates cross-country triple differences in trade data: Korea vs other countries, targeted vs non-targeted products, before/after 1973.

### 6.2 Equation (4a): DDD with flexible year interactions

With $c$ countries, $i$ products (SITC), $t$ years:

$$\begin{aligned} Y_{ict} = \alpha_i + \tau_t + \sigma_c + \sum_{j\neq 1972} \beta_{1,j}\big(\mathrm{HCI}_i \times \mathrm{Year}_t^j\big) + \sum_{j\neq 1972} \beta_{2,j}\big(\mathrm{Korea}_c \times \mathrm{Year}_t^j\big) \\ + \sum_{j\neq 1972} \beta_{3,j}\big(\mathrm{Korea}_c \times \mathrm{HCI}_i \times \mathrm{Year}_t^j\big) + \varepsilon_{ict}, \end{aligned} \tag{4a}$$

where $\mathrm{HCI}_i$ indicates products mapped to HCI-targeted industries. The object of interest is $\beta_{3,j}$.

### 6.3 Equation (4b): DDD with richer fixed effects

Lane also estimates a version with high-dimensional fixed effects:

$$Y_{ict} = \alpha_{ic} + \tau_{it} + \sigma_{ct} + \sum_{j\neq 1972} \beta_j\big(\mathrm{Korea}_c \times \mathrm{HCI}_i \times \mathrm{Year}_t^j\big) + \varepsilon_{ict}. \tag{4b}$$

### 6.4 Interpretation

The triple-interaction coefficients trace out whether Korea's targeted products increase comparative advantage *more than*:

- non-targeted Korean products (within-country difference), and
- targeted products in other countries (cross-country difference),

relative to 1972.

## 7 Model 4: Plant-level productivity comparisons

### 7.1 Production function setup (conceptual)

Plant-level TFP relies on a production function of the form:

$$y_{pt} = \beta_K k_{pt} + \beta_L \ell_{pt} + \omega_{pt} + \varepsilon_{pt},$$

where $\omega_{pt}$ is productivity observed by the firm but not the econometrician. Lane reports TFP estimates using several canonical methods (OP, LP, ACF, Wooldridge), each designed to address input endogeneity.

### 7.2 Equation (2): comparing TFP across targeted vs non-targeted plants

In the post-1979 plant microdata, the paper estimates:

$$\text{TFP}_{pit} = \alpha_{it} + \beta \, \text{Targeted}_p + \varepsilon_{pit}, \tag{2}$$

where $\alpha_{it}$ are industry-by-year fixed effects. **Interpretation:** $\beta$ captures whether plants in targeted industries are systematically more productive than non-targeted plants *within the same industry-year environment.*

## 8 Model 5: Learning-by-doing (industry and plant experience)

### 8.1 Equation (5): experience slopes differ by treatment

Lane studies whether targeted industries exhibit stronger learning-by-doing:

$$Y_{it} = \beta_1 \, \text{Experience}_{it} + \beta_2 \left( \text{Experience}_{it} \times \text{Targeted}_i \right) + \theta \, \text{Size}_{it} + \alpha_i + \tau_t + X_{it}'\phi + \varepsilon_{it}. \tag{5}$$

**Interpretation:**

- $\beta_1$ is the learning slope for non-targeted industries,
- $\beta_1 + \beta_2$ is the learning slope for targeted industries,
- outcomes $Y_{it}$ include prices, unit costs, and TFP.

# 9 Model 6: Production-network exposure (IO linkages) and spillovers

## 9.1 Direct-requirements matrix and linkage exposure

Let $\alpha_{ij}$ be the direct requirement (input share) from sector $j$ in the production of sector $i$ in the 1970 IO table.

**Backward linkages (dependence on targeted inputs).**

$$\text{BackwardLinkage}_i \equiv \sum_{j \in \text{HCI}} \alpha_{ij}. \tag{6a}$$

**Forward linkages (sales to targeted industries).**

$$\text{ForwardLinkage}_i \equiv \sum_{j \in \text{HCI}} \alpha_{ji}. \tag{6b}$$

**Total linkages via the Leontief inverse.** Let $\ell_{ij}$ be an element of the Leontief inverse, capturing total (direct + indirect) requirements. Then:

$$\text{TotalBackwardLinkage}_i \equiv \sum_{j \in \text{HCI}} \ell_{ij}, \qquad \text{TotalForwardLinkage}_i \equiv \sum_{j \in \text{HCI}} \ell_{ji}.$$

## 9.2 Equation (7): event study with linkage exposure

The dynamic specification becomes:

$$\log(y_{it}) = \alpha_i + \tau_t + \sum_{j \neq 1972} \gamma_j \left(\text{BackwardLinkage}_i \times \text{Year}_t^j\right) + \sum_{j \neq 1972} \delta_j \left(\text{ForwardLinkage}_i \times \text{Year}_t^j\right) + \varepsilon_{it}. \tag{7}$$

## 9.3 Interpretation

- $\delta_j$ captures whether *downstream* industries that rely more on targeted intermediates expand more after 1973.
- Estimating (7) on the *non-targeted-only sample* isolates spillovers from directly treated sectors (because treated industries are removed).

# 10 Interpreting each figure (what it shows)

## 10.1 Figure I: policy intensity — investment/tax wedge and directed credit

- **Panel A (effective marginal tax rate):** targeted industries experience a large, persistent reduction in the effective marginal tax rate during the HCI period, indicating a strong investment subsidy wedge.

- **Panels B–C (KDB lending):** Korea Development Bank lending (total and machinery lending) shifts sharply toward targeted industries during the drive, consistent with directed credit as a key policy instrument.

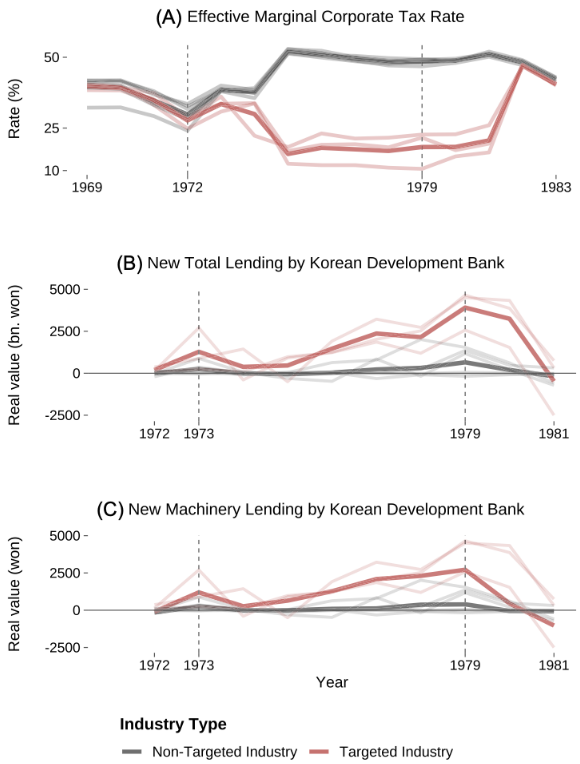


FIGURE I

Investment Policy and the Heavy and Chemical Industry Drive

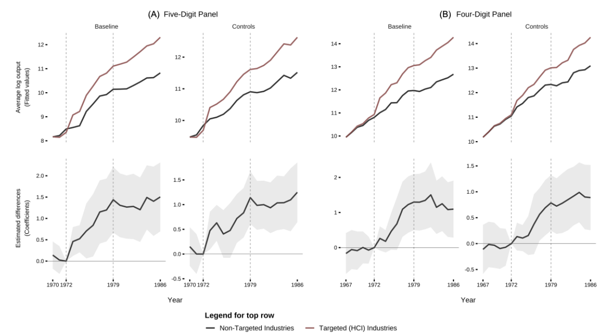


FIGURE II

Industrial Policy and Industry Output

## 10.2 Figure II: industrial output expands in targeted sectors (dynamic DiD)

Figure II plots the $\beta_j$ path from (1) for (log) real value shipped. Key reading:

- Pre-1973 coefficients are near zero $\Rightarrow$ no strong differential pre-trends.
- Post-1973 coefficients rise steadily $\Rightarrow$ targeted sectors expand relative to non-targeted.
- Effects persist beyond 1979 $\Rightarrow$ industrial expansion does not fully reverse when the drive ends.

## 10.3 Figure III: broader industrial and export development

**Panel A (industrial development).** Targeted sectors show:

- rising labor productivity,
- falling output prices,
- expanding employment and plant counts,

- increasing output and labor shares in manufacturing.

**Panel B (export development).** Comparative advantage measures (RCA and gravity-based proxies) increase gradually, with especially strong growth after 1973 and continued improvement after 1979.

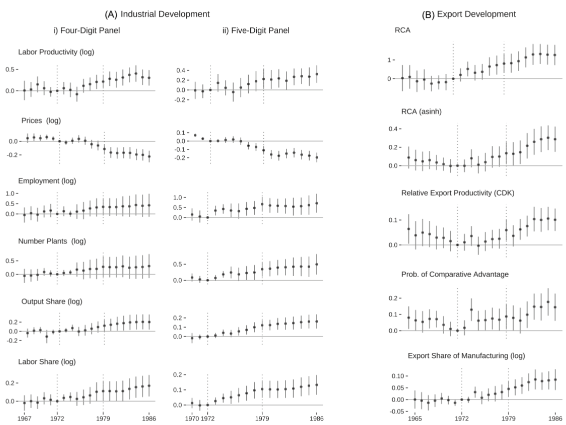


Figure III

Industrial Policy: Industrial and Export Development

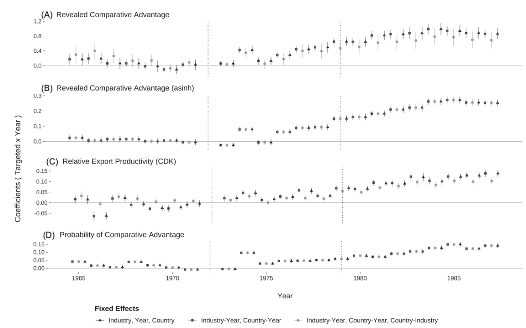


Figure IV

Cross-Country Triple Differences and Export Development

## 10.4 Figure IV: cross-country DDD confirms a Korea-specific targeted shift

Figure IV plots the Korea $\times$ Targeted $\times$ Year coefficients from (4a)–(4b). The post-1973 positive coefficients imply that Korea's targeted products upgrade in global markets beyond what is observed for:

- non-targeted Korean exports, and
- targeted exports in other countries.

## 10.5 Figure V: input use and investment increase (mechanism evidence)

Targeted industries increase intermediate input purchases, investment, and capital stock. This aligns with an *investment-policy* interpretation: the drive relaxed capital/credit constraints or subsidized capital accumulation in targeted sectors.

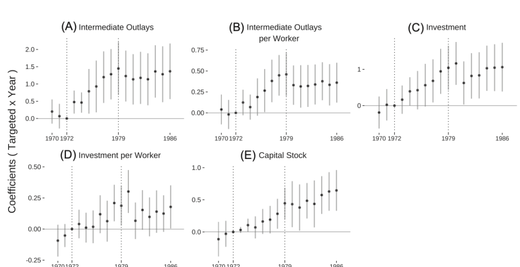


Figure V

Changes in Input Use and Investment

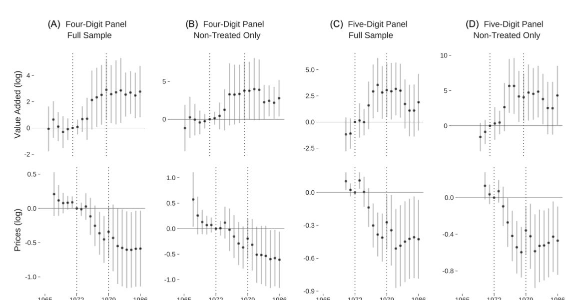


Figure VI

Forward-Linkages Exposure: Value Added and Output Prices

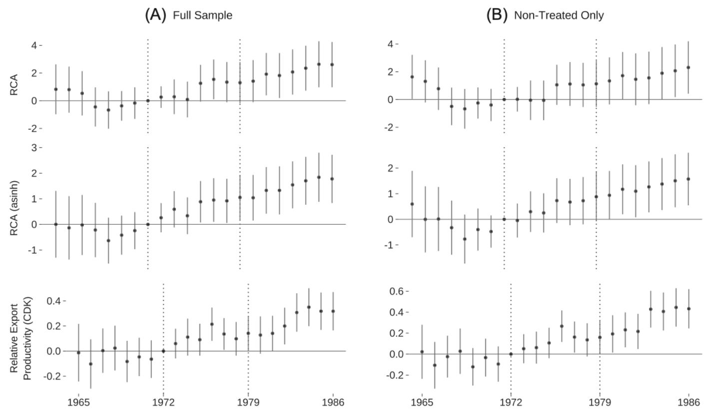


FIGURE VII

**Forward-Linkages Exposure: Export Development**

### 10.6 Figure VI: downstream (forward-linkage) spillovers on value added and prices

Industries that use targeted intermediates more intensively (higher forward-linkage exposure):

- expand value added after 1973,
- experience lower output prices (consistent with cost reductions / productivity improvements),

including in the non-targeted-only sample, supporting spillover effects.

### 10.7 Figure VII: downstream spillovers extend to export upgrading

Forward-linkage exposure also predicts improvements in export outcomes (RCA and relative export productivity) after 1973, again visible even among non-targeted industries.

## 11 Interpreting each table (what it implies)

### 11.1 Table I: post-1979 plant TFP is higher in targeted industries

Across several production-function estimators (OP/LP/ACF/Wooldridge), plants in targeted industries have higher TFP in 1980–1986, conditional on industry-year fixed effects. **Key takeaway:** productivity differences persist after the drive, consistent with long-run upgrading rather than a temporary stimulus.

### 11.2 Table II: average effects on industrial development are large and persistent

Table II reports average post-1973 effects using both doubly robust DiD and TWFE. Main patterns:

- **Output/value added/gross output:** large positive effects (e.g., log coefficients around 0.55–0.84, implying very large percentage increases).
- **Employment and plant counts:** positive expansion.
- **Prices:** negative coefficients (prices fall), consistent with productivity/efficiency gains.

TABLE I

DIFFERENCES IN PLANT-LEVEL TOTAL FACTOR PRODUCTIVITY, BY TREATMENT STATUS (1980–1986)

| | Outcomes: Total factor productivity | | | | |
|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) |
| Targeted | 0.043*** | 0.010 | 0.054*** | 0.042*** | 0.003 |
| | (0.015) | (0.014) | (0.015) | (0.014) | (0.013) |
| Industry × Year | Yes | Yes | Yes | Yes | Yes |
| $R^2$ | 0.368 | 0.446 | 0.342 | 0.234 | 0.117 |
| Observations | 272,150 | 272,150 | 272,150 | 272,150 | 272,150 |
| Two-way cluster (industry plant) | 488 × 91,094 | 488 × 91,094 | 488 × 91,094 | 488 × 91,094 | 488 × 91,094 |
| Estimation type (TFP) | W | ACF | LP | OP | OLS |

*Notes.* This table shows the relationship between plant-level TFP and HCI (targeted industries) for the post-HCI period (1980–1986). TFP is estimated using Ackerberg-Caves-Frazer (ACF), Levinsohn-Petrin (LP), Olley-Pakes (OP), and Wooldridge (W) methods.

TFP is estimated using a log-transformed, value-added production function. I include TFP estimated using OLS as a baseline estimate. The Targeted indicator is defined by the plant's main industry. All regressions control for year-by-industry (four-digit level) fixed effects. Regressions use two-way clustered standard errors at the plant and industry levels. Standard errors are in parentheses. * $p < .10$; ** $p < .05$; *** $p < .01$.

TABLE II

THE AVERAGE EFFECT OF INDUSTRIAL POLICY: INDUSTRIAL DEVELOPMENT

| | Five-digit panel | | Four-digit panel | |
|---|---|---|---|---|
| | Double robust | TWFE | Double robust | TWFE |
| Outcomes (log) | (1) | (2) | (3) | (4) |
| Output (shipm.) | 0.8378*** | 0.8235*** | 0.5923*** | 0.5452** |
| | (0.1764) | (0.1846) | (0.217) | (0.2223) |
| Value added | 0.7426*** | 0.7292*** | 0.5063** | 0.4586** |
| | (0.1696) | (0.1742) | (0.1973) | (0.209) |
| Gross output | 0.8383*** | 0.8236*** | 0.5962*** | 0.5481** |
| | (0.1718) | (0.1852) | (0.2033) | (0.2217) |
| Employment | 0.504*** | 0.4972*** | 0.2941 | 0.2679 |
| | (0.1451) | (0.1509) | (0.204) | (0.1915) |
| Prices | −0.1002*** | −0.1012*** | −0.1154*** | −0.1152*** |
| | (0.0207) | (0.0205) | (0.0305) | (0.0304) |
| Labor prod. | 0.1608** | 0.1548** | 0.1602** | 0.1371* |
| | (0.0654) | (0.068) | (0.0688) | (0.0829) |
| Output share | 0.0996*** | 0.0993*** | 0.1072* | 0.097 |
| | (0.0258) | (0.0261) | (0.0551) | (0.0599) |
| Labor share | 0.0979*** | 0.0967*** | 0.1254** | 0.116** |
| | (0.0284) | (0.028) | (0.0534) | (0.0495) |
| Num. plants | 0.297*** | 0.2908*** | 0.1986 | 0.1831 |
| | (0.0977) | (0.1018) | (0.1419) | (0.1549) |

*Notes.* This table shows the ATT for industrial policy. Average DD estimates are shown for double-robust and TWFE estimators. Outcomes are log: output is the real value of gross output shipped (shipments), alongside other measures of real output: value added and gross output. Employment is the total number of workers. Prices are industry output prices. Labor prod. is real value added per employee. Output share is the manufacturing share of industry output. Labor share is the manufacturing share of industry employment. Specifications include controls for pre-1973 industry averages (log): average wages, average plant size, intermediate outlays, and labor productivity. Standard errors, clustered at the industry level, are in parentheses. Double-robust DD estimates come from equation (3). Double-robust estimators use bootstrapped standard errors (10,000 iterations) and are adjusted to allow for within-industry correlation. * $p < .10$; ** $p < .05$; *** $p < .01$.

- **Shares:** targeted industries' output and labor shares in manufacturing increase, consistent with reallocation toward targeted sectors.

### 11.3 Table III: export development improves in targeted sectors

Targeted industries:

- increase Balassa RCA and transformed RCA measures,
- become more likely to exceed RCA $> 1$ (comparative advantage),
- increase their export share of manufacturing,
- show improvements in gravity-based relative export productivity (CDK).

**Interpretation:** the policy coincides with a gradual but durable shift in dynamic comparative advantage.

### 11.4 Table IV: inputs and investment surge in targeted industries

Targeted industries raise intermediate input usage and investment (and per-worker measures), consistent with policy-driven capital deepening and expansion of production capacity.

### 11.5 Table V: industry-level learning-by-doing is stronger in targeted industries

Table V estimates (5) at the industry level. Interpretation of signs:

- **Prices:** experience tends to reduce prices; targeted $\times$ experience is additionally negative $\Rightarrow$ faster price declines in targeted sectors.
- **Unit costs:** with richer controls, experience reduces unit costs; targeted $\times$ experience is more negative $\Rightarrow$ steeper cost reductions for targeted sectors.

TABLE III

THE AVERAGE EFFECT OF INDUSTRIAL POLICY: EXPORT DEVELOPMENT

| | Type of Estimator | | |
|---|---|---|---|
| Outcomes | Double robust (1) | TWFE (2) | PPML (3) |
| RCA | 0.4853*** | 0.4701*** | 0.9142*** |
| | (0.1841) | (0.1806) | (0.26) |
| RCA (log) | 0.1251*** | 0.1192*** | 0.5939*** |
| | (0.0419) | (0.042) | (0.1535) |
| RCA (asinh) | 0.1633*** | 0.1557*** | 0.6059*** |
| | (0.0513) | (0.0537) | (0.1577) |
| RCA (CDK) | 0.0502*** | 0.0498*** | 0.0302 |
| | (0.0182) | (0.0176) | (0.0189) |
| Prob. comparative adv. | 0.1057*** | 0.1021*** | 0.6486*** |
| | (0.0281) | (0.0307) | (0.1945) |
| Export share | 0.071** | 0.0727** | 0.8346*** |
| | (0.0299) | (0.0293) | (0.2582) |
| Export share (log) | 0.0481*** | 0.048*** | 0.7658*** |
| | (0.0145) | (0.0147) | (0.1991) |
| Export share (asinh) | 0.0596*** | 0.0599*** | 0.7998*** |
| | (0.0196) | (0.0191) | (0.2166) |

*Notes*. This table shows the ATT for industrial policy. Average DD estimates are shown for double-robust, PPML TWFE, and linear TWFE estimators. RCA is the standard Balassa index measure of revealed comparative advantage. RCA (CDK) is relative productivity estimated using CDK. See the text for their calculation. The indicator $\mathbb{1}[\text{RCA} > 1]$ is a binary dummy variable equal to one when an industry has achieved comparative advantage, zero otherwise. I also show transformed versions of RCA (asinh and log). Specifications include controls for pre-1973 industry averages (log): avg. wages, avg. plant size, intermediate outlays, and labor productivity. Standard errors, clustered at the industry level, are in parentheses. Double-robust DD estimates come from equation (3). Double-robust estimators use bootstrapped standard errors (10,000 iterations) and are adjusted to allow for within-industry correlation. * $p < .10$; ** $p < .05$; *** $p < .01$.

TABLE IV

THE AVERAGE EFFECT OF INDUSTRIAL POLICY: INPUT USE AND INVESTMENT

| | Five-digit panel | | Four-digit panel | |
|---|---|---|---|---|
| Outcomes (log) | Double robust (1) | TWFE (2) | Double robust (3) | TWFE (4) |
| Intermediate outlays | 0.7544*** | 0.7408*** | 0.5606** | 0.5147** |
| | (0.1878) | (0.1894) | (0.2374) | (0.2428) |
| Intermediate outlays (per worker) | 0.213*** | 0.2074*** | 0.2823*** | 0.2659*** |
| | (0.069) | (0.0719) | (0.0923) | (0.0975) |
| Investment | 0.6198*** | 0.6089*** | 0.2445 | 0.21 |
| | (0.2211) | (0.2205) | (0.2136) | (0.2186) |
| Investment (per worker) | 0.134** | 0.1316** | 0.1124 | 0.1048 |
| | (0.0596) | (0.0612) | (0.0855) | (0.0901) |

*Notes*. This table shows the ATT for industrial policy. Average DD estimates are shown for double-robust and TWFE estimators. Intermediate outlays (log) are real intermediate input costs. Investment total (log) is real total gross capital formation. I also show outcomes in per worker terms. Specifications include controls for pre-1973 industry averages (log): average wages, average plant size, intermediate outlays, and labor productivity. Standard errors, clustered at the industry level, are in parentheses. Double-robust DD estimates come from equation (3). Double-robust estimators use bootstrapped standard errors (10,000 iterations) and are adjusted to allow for within-industry correlation. * $p < .10$; ** $p < .05$; *** $p < .01$.

- **TFP:** experience increases TFP; targeted × experience is positive in several specifications ⇒ faster productivity growth with experience in targeted sectors.

**Mechanism reading:** consistent with infant-industry dynamics and learning effects rather than purely static reallocations.

TABLE V

INDUSTRY-LEVEL LEARNING BY TREATMENT STATUS

| | | | | | Total factor productivity | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Prices (log) | | Unit cost (log) | | ACF | | LP | | W | |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) |
| Experience | −0.011 | −0.195*** | 0.015*** | −0.109*** | 0.182*** | 0.369*** | 0.053 | 0.360*** | 0.060 | 0.348*** |
| | (0.008) | (0.029) | (0.004) | (0.014) | (0.065) | (0.059) | (0.083) | (0.060) | (0.081) | (0.064) |
| Targeted × Experience | −0.042*** | −0.044*** | 0.005 | −0.035*** | 0.014 | 0.033 | 0.087** | 0.120*** | 0.094*** | 0.125*** |
| | (0.008) | (0.012) | (0.007) | (0.009) | (0.031) | (0.022) | (0.039) | (0.025) | (0.036) | (0.024) |
| Controls for size-scale | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Controls for capital intensity | No | Yes | No | Yes | No | Yes | No | Yes | No | Yes |
| Controls for intermediates | No | Yes | No | Yes | No | Yes | No | Yes | No | Yes |
| Controls for investment | No | Yes | No | Yes | No | Yes | No | Yes | No | Yes |
| Industry effects | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Year effects | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| $R^2$ | 0.951 | 0.961 | 0.845 | 0.900 | 0.823 | 0.877 | 0.976 | 0.985 | 0.983 | 0.990 |
| Observations | 3,890 | 3,429 | 3,890 | 3,429 | 3,512 | 3,428 | 3,512 | 3,428 | 3,512 | 3,428 |
| Clusters | 278 | 263 | 278 | 263 | 264 | 263 | 264 | 263 | 264 | 263 |
| Linear combination | −0.053 | −0.239 | 0.020 | −0.143 | 0.196 | 0.402 | 0.140 | 0.480 | 0.154 | 0.474 |
| (std. err.) | (0.009) | (0.029) | (0.007) | (0.016) | (0.060) | (0.054) | (0.073) | (0.059) | (0.072) | (0.062) |

*Notes*. This table shows the industry-level relationship between industrial outcomes and (log) experience in targeted versus non-targeted industries. Estimates come from equation (5). The analysis is for the post-1972 period, using the five-digit industry panel. The outcomes are log unit cost (total intermediate costs per unit of real gross output) and TFP, estimated using Ackerberg-Caves-Frazer (ACF), Levinsohn-Petrin (LP), and Wooldridge (W) methods. (log) Experience is measured as cumulative output (the sum of real gross output until the current year). All equations control for size/scale, measured as (log) industry employment and (log) average plant size. Additional controls include log: capital intensity, investment per worker, and intermediate input intensity per worker. Linear combination, at the bottom, gives the combined effects. All specifications are estimated using industry and year fixed effects. Standard errors, clustered at the industry level, are in parentheses. * $p < .10$; ** $p < .05$; *** $p < .01$.

TABLE VI

PLANT- AND INDUSTRY-LEVEL LEARNING BY TREATMENT STATUS

| | Unit cost (log) | | | TFP | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Plant experience | −0.072*** | −0.072*** | −0.069*** | 0.479*** | 0.485*** | 0.484*** |
| | (0.002) | (0.002) | (0.002) | (0.011) | (0.011) | (0.011) |
| Targeted × plant experience | −0.009*** | −0.008*** | −0.005*** | 0.032*** | 0.014* | 0.018** |
| | (0.001) | (0.001) | (0.001) | (0.007) | (0.008) | (0.007) |
| Industry experience | | −0.008*** | −0.006** | | 0.022 | 0.026* |
| | | (0.003) | (0.003) | | (0.014) | (0.015) |
| Targeted × industry experience | | −0.002** | −0.003*** | | 0.030*** | 0.030*** |
| | | (0.001) | (0.001) | | (0.009) | (0.009) |
| Control for plant size | Yes | Yes | Yes | Yes | Yes | Yes |
| Control for capital | Yes | Yes | Yes | Yes | Yes | Yes |
| Control for skill ratio | Yes | Yes | Yes | Yes | Yes | Yes |
| Control for investment | Yes | Yes | Yes | Yes | Yes | Yes |
| Control for intermediates | Yes | Yes | Yes | Yes | Yes | Yes |
| Polynomial controls | No | No | Yes | No | No | Yes |
| Plant effect | Yes | Yes | Yes | Yes | Yes | Yes |
| Year effect | Yes | Yes | Yes | Yes | Yes | Yes |
| Industry effect | Yes | Yes | Yes | Yes | Yes | Yes |
| $R^2$ | 0.882 | 0.882 | 0.890 | 0.691 | 0.691 | 0.694 |
| Observations | 250,989 | 250,989 | 250,989 | 235,940 | 235,940 | 235,940 |
| Clusters (industry and plant) | 489 × 60,009 | 489 × 60,009 | 489 × 60,009 | 489 × 57,942 | 489 × 57,942 | 489 × 57,942 |

TABLE VI

CONTINUED

| | Unit cost (log) | | | TFP | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Linear combination (plant level) | −0.081 | −0.080 | −0.073 | 0.510 | 0.499 | 0.502 |
| (std. err.) | (0.002) | (0.002) | (0.002) | (0.013) | (0.013) | (0.013) |
| Linear combination (industry level) | | −0.010 | −0.009 | | 0.052 | 0.056 |
| (std. err.) | | (0.003) | (0.003) | | (0.014) | (0.015) |

*Notes*. This table shows the plant-level relationship between industrial development and (log) experience in targeted versus non-targeted industries. Estimates come from a plant-level version of equation (5). Outcomes are the following: log unit cost (total intermediate costs per unit of real gross output) and TFP (estimated using Ackerberg-Caves-Frazer). Experience is measured as cumulative output (the sum of real gross output until the current year). Plant experience refers to plant-level cumulative learning, and industry experience refers to industry-level learning, calculated at the four-digit industry level. All equations control for log plant size (employment). Additional controls include log: capital intensity, skill ratio, investment per worker, and intermediate input intensity per worker. Linear combination, at the bottom, gives the combined effects. All specifications are estimated using plant, industry, and year fixed effects. Polynomial controls adds cubic polynomials in the control variables. Two-way standard errors are clustered at the industry and plant levels and are in parentheses. * $p < .10$; ** $p < .05$; *** $p < .01$.

## 11.6 Table VI: learning decomposes into within-plant and across-plant components

Table VI adds plant experience and industry experience simultaneously. Key takeaway:

- **Within-plant learning:** more experience predicts lower unit costs and higher TFP; slopes are steeper for targeted plants.

- **Industry-wide spillovers:** industry experience also matters (especially for TFP), suggesting learning externalities across plants within an industry.

## 11.7 Table VII: forward-linkage exposure predicts post-1973 value-added growth

Industries with stronger downstream exposure to targeted intermediates expand more after 1973. The effect remains when focusing only on non-targeted industries, supporting network spillovers beyond directly treated sectors.

TABLE VII

LINKAGE EXPOSURE AND VALUE ADDED, BEFORE AND AFTER 1973

| | Outcome: Value added (log) | | | |
|---|---|---|---|---|
| | Five-digit panel (1970–1986) | | Four-digit panel (1967–1986) | |
| | Full sample (1) | Non-HCI sample (2) | Full sample (3) | Non-HCI sample (4) |
| Post × forward linkage | 2.832*** | 4.405*** | 2.095** | 2.906** |
| | (0.914) | (1.504) | (0.802) | (1.174) |
| Post × backward linkage | −0.0167 | 0.176 | −0.693 | −2.163* |
| | (0.334) | (0.375) | (0.559) | (1.279) |
| Industry effects | Yes | Yes | Yes | Yes |
| Year effects | Yes | Yes | Yes | Yes |
| Targeted × year | Yes | No | Yes | No |
| $R^2$ | 0.776 | 0.763 | 0.847 | 0.819 |
| Observations | 4,720 | 2,986 | 1,750 | 1,096 |
| Clusters | 278 | 176 | 88 | 55 |

*Notes*. Average DD estimates, before and after 1973. Estimates correspond to equation (7). Regressions interact linkage measures with a Post indicator. The outcome is real log value added. Both linkage interactions (forward and backward) are shown. Analysis is performed for the sample of only non-treated industries and the full sample of industries. Estimates for the full sample separately control for the Targeted × Year effects to account for the main effect of policy. Standard errors, clustered at the industry level, are in parentheses. * $p < .10$; ** $p < .05$; *** $p < .01$.

TABLE VIII

LINKAGE EXPOSURE AND OUTPUT PRICES, BEFORE AND AFTER 1973

| | Outcome: Output prices (log) | | | |
|---|---|---|---|---|
| | Five-digit panel (1970–1986) | | Four-digit panel (1967–1986) | |
| | Full sample (1) | Non-HCI sample (2) | Full sample (3) | Non-HCI sample (4) |
| Post × forward linkage | −0.359*** | −0.459*** | −0.483** | −0.510*** |
| | (0.128) | (0.144) | (0.184) | (0.176) |
| Post × backward linkage | 0.103*** | 0.0880*** | 0.251 | 0.673*** |
| | (0.0213) | (0.0142) | (0.154) | (0.226) |
| Industry effects | Yes | Yes | Yes | Yes |
| Year effects | Yes | Yes | Yes | Yes |
| Targeted × year | Yes | No | Yes | No |
| $R^2$ | 0.957 | 0.942 | 0.962 | 0.956 |
| Observations | 4,721 | 2,987 | 1,751 | 1,097 |
| Clusters | 278 | 176 | 88 | 55 |

*Notes*. Average DD estimates, before and after 1973. Regressions interact linkage measures with a Post indicator. Estimates correspond to equation (7). The outcome variable is log output price. Both linkage interactions (forward and backward) are shown. Analysis is performed for the sample of only non-treated industries and the full sample of industries. Estimates for the full sample separately control for the Targeted × Year effects to account for the main effect of policy. Standard errors, clustered at the industry level, are in parentheses. * $p < .10$; ** $p < .05$; *** $p < .01$.

### 11.8 Table VIII: forward-linkage exposure predicts lower post-1973 output prices

Higher exposure to targeted intermediates is associated with lower output prices after 1973, consistent with cheaper/better inputs and cost reductions propagating downstream.

## 12 Synthesis: what the paper ultimately argues

The empirical evidence supports a coherent narrative:

1. **Direct sectoral expansion:** targeted industries grow sharply in output and factor usage during the HCI drive and remain larger afterward.
2. **Dynamic upgrading:** export comparative advantage shifts gradually and persists beyond the intervention window.
3. **Learning mechanisms:** targeted sectors show stronger learning-by-doing, with both within-plant and industry-level spillovers.
4. **Network spillovers:** downstream sectors with stronger input linkages to targeted industries grow and become more competitive, consistent with production-network propagation.

## 13 Conclusion

### 13.1 Summary

Lane (2025) concludes that South Korea's 1973–1979 Heavy and Chemical Industry (HCI) drive *causally* promoted industrialization in the manufacturing sectors targeted by the policy, and that the effects were **persistent**, **multidimensional**, and **propagated through production networks**.

The paper's central conclusion has three layers that fit together:

1. **Direct industrial development:** targeted industries expand in real activity (output/value added/employment/plants) and become more competitive (lower output prices and higher

productivity proxies). These gains are not a short-lived "stimulus" effect; they remain visible even after the major HCI interventions were retrenched.

2. **Dynamic export upgrading:** targeted industries gradually improve their export performance and comparative advantage. Importantly, export effects are *slow-moving* and in some cases become strongest *after* the formal policy window ends, consistent with capability accumulation rather than an immediate demand or protection effect.

3. **Spillovers beyond treated sectors:** the HCI drive also affects non-targeted industries through input–output linkages. Downstream industries that rely more on HCI intermediates expand more and experience lower output prices, indicating that the policy operated partly by shifting the cost and availability of key intermediate inputs.

Overall, the paper argues that the episode looks less like "miraculous" industrialization and more like a combination of **conventional policy forces**: investment subsidies (lower user cost wedges), directed credit, and an emphasis on upstream intermediates, which then interact with learning-by-doing and network propagation.

### 13.2 Key quantitative takeaways

Using the preferred average double-robust DiD estimates in the five-digit panel (Table II), the average treatment effects imply economically large changes for targeted industries (relative to non-targeted):

- **Real output (shipments):** $\widehat{\beta} \approx 0.84$ log points $\Rightarrow$ about $e^{0.84} - 1 \approx 131\%$ higher.
- **Real value added:** $\widehat{\beta} \approx 0.74$ log points $\Rightarrow$ about 110% higher.
- **Employment:** $\widehat{\beta} \approx 0.50$ log points $\Rightarrow$ about 66% higher.
- **Output prices:** $\widehat{\beta} \approx -0.10$ log points $\Rightarrow$ about 9.5% lower.

Export upgrading (Table III) is also sizable:

- **Probability of achieving comparative advantage:** Pr(RCA $> 1$) rises by about 0.106 (roughly 10 percentage points).
- **Export share:** increases by about 0.071 (about 7 percentage points in the reported share metric).

These magnitudes complement the dynamic event-study evidence: the post-1973 gap grows over time and does not quickly revert after 1979.

### 13.3 Economic intuition: why policy effects persist and why exports respond with a lag

A coherent mechanism that rationalizes the main patterns is:

$$\begin{aligned}\text{policy wedges (tax/credit)} \;\Rightarrow\; &\text{investment \& scale} \;\Rightarrow\; \text{learning and cost reductions}\\ \Rightarrow\; &\text{export competitiveness} \;\Rightarrow\; \text{network spillovers.}\end{aligned}$$

**1) Investment wedge $\Rightarrow$ capacity and scale.** The HCI drive lowered the effective cost of investing (tax and financing wedges) and expanded directed credit to targeted sectors. This raises capital accumulation and intermediate input use in treated industries, allowing them to operate at larger scale.

**2) Scale $\Rightarrow$ learning-by-doing and capability accumulation.** If unit costs fall with cumulative experience (at the plant level and via within-industry spillovers), then a temporary policy that pushes an industry onto a higher-production path can generate *persistent* productivity gains. Conceptually, a reduced-form learning relationship can be written as:

$$\log c_{it} = \kappa_i + \lambda_t - \phi \cdot \text{Experience}_{it} - \phi_T \cdot \big(\text{Experience}_{it} \times \text{Targeted}_i\big) + \cdots ,$$

so targeted industries have a steeper experience–cost gradient (larger $\phi_T$), which produces sustained price declines.

**3) Why export upgrading is slow-moving.** Export comparative advantage depends on more than contemporaneous costs; it requires time to build:

- product quality and reliability,
- production know-how and supply-chain coordination,
- buyer relationships and market access.

Hence, even if domestic output expands quickly during 1973–1979, export performance can improve gradually and materialize strongly later, which is exactly what the paper emphasizes.

**4) Why downstream spillovers arise (the network channel).** When treated sectors are upstream intermediates, they enter other sectors' marginal costs. A simple downstream cost channel is:

$$c_{it} = c\big(w_t, r_t, \{p_{jt}\}_{j \in \mathcal{I}_i}\big), \qquad \frac{\partial c_{it}}{\partial p_{jt}} > 0,$$

so if the policy lowers the prices (or improves the quality) of HCI inputs $p_{jt}$, downstream costs fall. Downstream industries then expand and can become more competitive, including in exports. This is why the paper finds the strongest indirect effects via *forward* linkages to treated upstream sectors.

### 13.4 Takeaway

*Lane (2025) concludes that Korea's HCI drive generated large and persistent industrialization in targeted sectors, with export comparative advantage upgrading that emerges gradually and spillovers that propagate downstream through intermediate-input linkages—consistent with industrial policy working through investment wedges, learning-by-doing, and production networks.*